

Enterprise Resource Planning System - Technical Documentation

Document Type: Documentation

System Overview

The Enterprise Resource Planning (ERP) System is a comprehensive software solution designed to integrate and manage core business processes. This documentation provides detailed information about system architecture, configuration, and usage.

Architecture

The system follows a three-tier architecture: presentation layer (web interface), business logic layer (application server), and data layer (database). Built using Java Enterprise Edition with Oracle database backend. Supports multiple deployment models including on-premise and cloud.

Installation and Setup

System requirements: Java 11+, Oracle Database 19c, 16GB RAM minimum. Installation process involves database setup, application server configuration, and web server deployment. Detailed setup scripts and configuration files are provided in the installation package.

User Management

Role-based access control with predefined user roles: Administrator, Manager, Employee, and Viewer. User authentication supports LDAP integration and multi-factor authentication. Password policies enforce security requirements and regular updates.

Module Configuration

Core modules include: Financial Management, Human Resources, Inventory Management, and Customer Relationship Management. Each module can be configured independently with specific business rules and workflows. Custom fields and reports can be added as needed.

Database Schema

The database consists of 150+ tables organized into functional schemas. Primary tables include users, organizations, transactions, and audit logs. Foreign key relationships maintain data integrity across modules. Indexing strategies optimize query performance.

API Documentation

RESTful API endpoints provide integration capabilities with external systems. Authentication uses OAuth 2.0 with JWT tokens. API responses follow standard HTTP status codes and JSON format. Rate limiting and throttling prevent abuse.

Troubleshooting

Common issues include database connection problems, memory allocation errors, and configuration conflicts. Diagnostic tools provide detailed error logging and system health monitoring. Support procedures include escalation paths and resolution timeframes.

Maintenance

Regular maintenance tasks include database backups, log rotation, and performance monitoring. Patch management follows a quarterly release schedule. System updates require scheduled downtime and rollback procedures are documented.